Computer Networks

google.com $\xrightarrow{\text{request}}$ Server
google.com $\xleftarrow{\text{response}}$ Server

Client — Server

↳ Our own computer can act as a client & server when we work on local host.

↳ Protocols are the rules defined by the internet society.

TCP - Transmission control protocol it ensures the data reaches the destination without being corrupted along the way.

UDP - User Data Protocol when you dont care if 100% of your data is reaching the destination. For ex: Video conferencing

HTTP - Hyper text transfer protocol defines the format of the data that is being transferred used by the world wide web. Client - Server req-response are given by the HTTP, how server sends back the data is a rule in HTTP.

↳ Data received will be in Packets, when u reload a page we get a lot of get & post request on the network tab which are information sent in small chunks of packets.

↳ IP address is just like a phone book where phone numbers are matched with names similarly the IP addresses are matched with domains.

google.com → $\underbrace{X}_{0-255} . X . X . X$ → format of IP

To check IP address of your computer
$ curl if config.me -S

↳ ISP : Internet service provider will provide modem/ or router and devices connected with it will have same local IP address. It does it by DHCP (Dynamic host control protocol) it is set of rules & regulations

Internet (google.com) ⟷ ISP ⟷ Modem/Router (It has global IP) → D_1 IP_1 If we hit google.com from D_1, D_2, D_3 it will use global IP and shows modem is requesting
→ D_2 IP_2
→ D_3 IP_3

Say D_1 requests for google.com it passes modem goes to ISP fetches google from internet, returns to modem and then modem decides who gave the request say D_1 and it decides by using NAT (Network access translator)

↳ Now lets say modem knows D_1 has requested but to which application we must send the data. As all the device will have same IP address?

⇒ we do it using port numbers, IP locates which device and port decides which application

↳ Port is a 16 byte number, total port numbers that you can create is $2^{16} \approx 65{,}000$ port no. are available.

↳ All the HTTP stuff is on port 80 well defined similarly mongodb is on port 27017.
Ports from 0-1023 are reserved ports

↳ So it means u cannot host your application on port 80 as it is reserved by HTTP.

↳ Port from 1024 - 49152 are reserved for applicat^n
Ex SQL runs on port 1433
Remaining we can use it for our connection

⇒ 1mbps = 1 mega bits per second i.e 10^6 bits/s
1gbps = 10^9 bits/s
1Kbps = 10^3 bits/s

↳ The computers are connected physically with the help of wires and networks which are laid under the ocean, those are optical fibre cables, coaxial cables. Faster than satellites.

↳ Wireless stuff consists of bluetooth, wifi (short ranges), 3G, 4G, 5G for longer ranges.

↳ LAN : Local area network for small house/office
we can connect with Ethernet, wifi.

↳ MAN : Metropolitan are network which spans across the city.

↳ WAN : Wide area networks accross country.
Basically internet is a collection of all these 3.

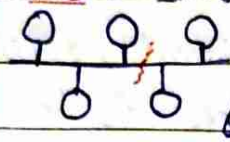
⇒ SONET : Synchronous optical networking carries the data using optical fibres and can cover distance

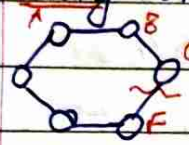
⇒ Frame relay : A way to connect local area network (LAN) to wide area network (WAN)

↳ Modem is used to convert digital signals to analog signals and vice versa. Digital data on PC converts to electric signals by modem and can transfer these signals, and also recover the digital data.

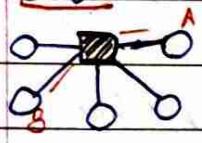
↳ **Router** :- It routes the data packets packages based on their IP address, Study more in OSI
We have few Tier 1, ISP in India we have Tata whereas Tier 2 has airtel, vodaphone etc

↳ **Topologies** :-

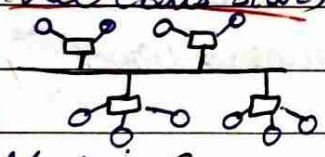
i) **Bus** : If one of the links get broken it will disrupt the entire network.

Only 1 person can send data at a time.

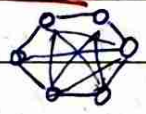
ii) **Ring** : Every system communicates with one another


goes through every computer calls and if one breake the network distrupts.

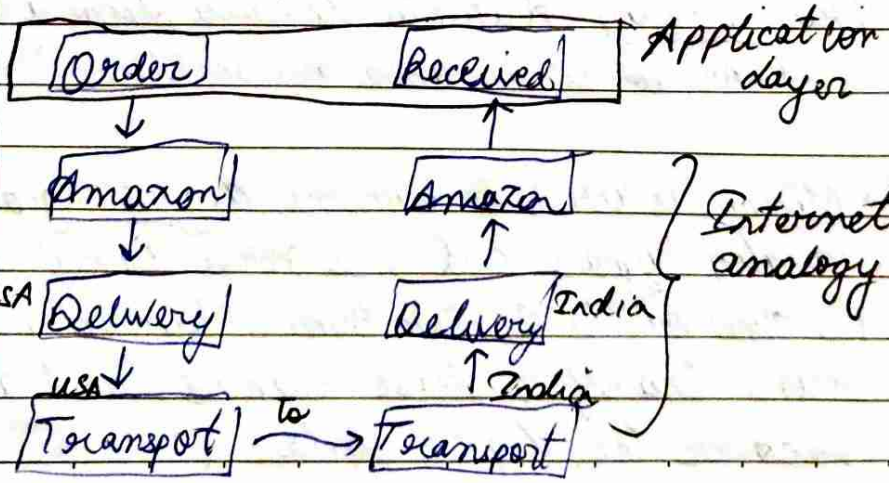
iii) **Star** : All computers are connected to a


controlling device. If center fails computer network goes down.

iv) **Tree (Bus-star)** : Many stars are connected

in a bus topology. Has more fault tolerance.

v) **Mesh** :- Every computer is connected to each other,

it is expensive, scalability issue.

★ **Structure of Network** :- OSI model



OSI : open systems interconnected model is a standard way of how 2 or more computers interact with each other. There are 7 layers in OSI model.

i) Application : Users interact with it software/application.

ii) Presentation : Takes data from application in number, string and converts into machine readable binary format. Ex it converts it from ASCII to EBCDIC. It also encodes & encrypts the data such thats only end user can read that data. Also provides abstraction, also compresses the data. SSL sockets are used.

iii) Session : Helps in setting up and managing the connections and it enables sending and receiving data followed by terminated of connected sessions. Like authentication types thing.

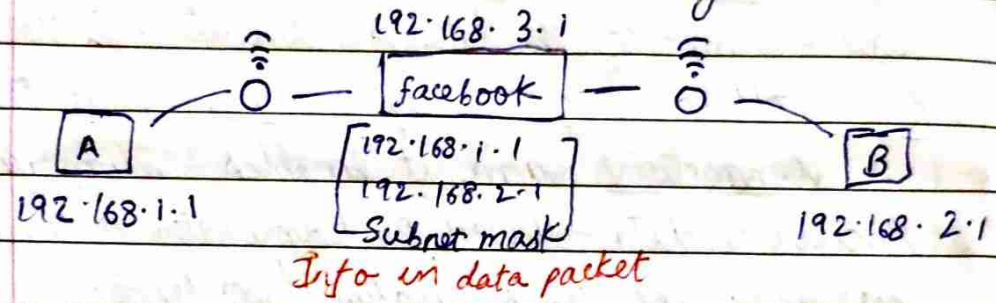
iv) Transport : Sends the data using TCP, UDP protocols. From session data is divided into segments and each segment has source & end portnumber and sequence no. to regain the data packets. Flow control layer controls the amount of data transferred like user has 40mbps & reciever has 20mbps so it slows. It adds check sum to every data segment good or not

v) Network : Transmission of recieved data segments from one computer to another located at different network host. This is where router lives. Logical addressing assigns sender & reciever IP address to every data packet. Performs routing moving packets from src to destination

vi) Data Links : Recieves data from network and does physical addressing. Creates frames and data units are passed. It uses mac address for that.

Mac address is a 12-digit alphanumeric no of the network interface of your computer. Computer's bluetooth, wifi etc may have different mac address. Data link layer adds mac address in a frame in a packet and pushes and transports that frame.

vii) Physical: Contains the hardware section, like wires and transmits the bits. It recieves the bits converts it at reciever makes frames and passes it to the data layer.



User 1		User 2
Applicat^n	⟷	Application
Presentat^n	⟷	Presentation
Session		Session
Transport		Transport
Network		Network
Data Layer	⟷	Data Layer
Physical	—	Physical

Each layer imagines that they are talking to there opp. layer of user 1 & 2. Internally it takes this route.

★ TCP/IP model: Internet protocol suit (5 layers)

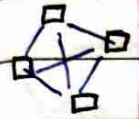
Application	Applicat^n, Present^n & Session are merged in one layer. Used commonly
Transport	
Network	
Data link	
Physical	

1. Application Layer: Users interact with it.
It lies on your devices.

a) Client sends request and servers send response.
Server is a system which controls the hosting website.
Collection of servers is known as data center.
ping google.com It sends request to google.com
64 bytes from 142.80.8 icmp-seq=1 ttl=60 time=17.8ms
↳ Bytes passed ↳ IP ↳ Packet seq ↳ Time to live ↳ Time taken

Round trip time from host to destinat^n computer and echoed back is known as ping time.

b) Peer to peer architecture (P2P): Applications running on various devices they get connected with each other. No server or datacenter
For ex computers connected in college. we can scale it rapidly, decentralized network.
Every computer can be client as well as server.



↳ Protocols: There are various protocols for different tasks associated to. lets look at web protocols
TCP/IP: HTTP (for web), DHCP (Packets), FTP (File transfer), SMTP (send mail), POP3 & IMAC (recive email)

SSH: To login into someone else terminal
VNC: For graphical control (virtual network computing)
Telnet: Terminal emulation enables user to connect to host or device using Telnet client (Port 23)
UDP: Stateless connection, data may be lost.

Program WhatsApp
Process: send message, record video
Threads: set up page, open camera

↳ Sockets: It is the interface that is b/w the processes and the internet know as socket

↳ Ports: Say you need to send response to chrome but there are many tabs opened on chrome, so we use Ephemeral ports.
Say chrome works on port 80 it will internally assign random port numbers; If multiple instances are running once done it is freed.
Ephemeral ports they can exist on the client side but on the server you must know the port number*

↳ HTTP: It is a client server protocol and it tells how you request this data from the server and also how the server will send back data to the client.
These are application layer protocols; every application layer protocol also require transport layer protocol a) HTTP uses TCP protocol to send-receive data.
(b) HTTP is a stateless protocol; server will not store any information of client by default

↳ HTTP Methods:
GET: Requesting some data
POST: Giving something to server
PUT: Puts data at specific location
DELETE: Delete something from the server.

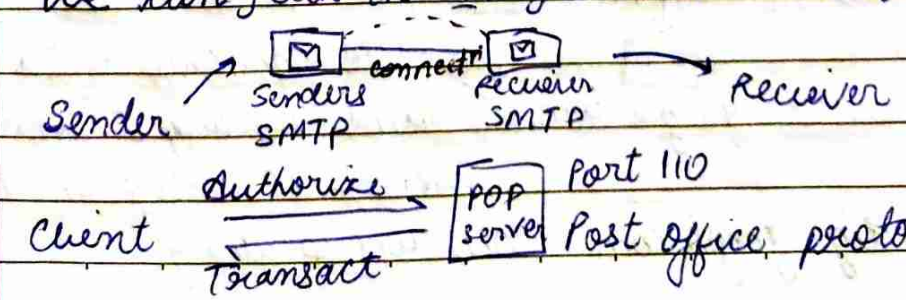
↳ Status codes

1xx → Informational category
2xx → Success codes
3xx → Redirection purpose
4xx → Client Error
5xx → Server Error

↳ Cookies : It is a unique string, and it is stored on the clients (our) browser.
If u visit a site first time it will set a cookie, and when you revisit the site it will check the state of your cookie.
When you make new request, cookie will be send in the header. Server know this cookies comes from say Sahu and know its state

↳ Third party cookies : Cookies that are set for the urls that you do not visit.
like used for adsense on ecom sites.

↳ How email works : SMTP simple mail transfer protocol, to send email. To recieve email we use POP3 at the application layer protocol. It uses TCP protocol.
nslookup -type=mx gmail.com.
We can fetch the IP of the SMTP server



↳ IMAP - Internet message access protocol used to get emails and allows you to view your emails on multiple devices. local copies will be available from various devices

↳ DNS : Domain Name Systems are mapped to the IP address, it is part of the application layer HTTP converts the domain name to IP and finds the IP and returns it.

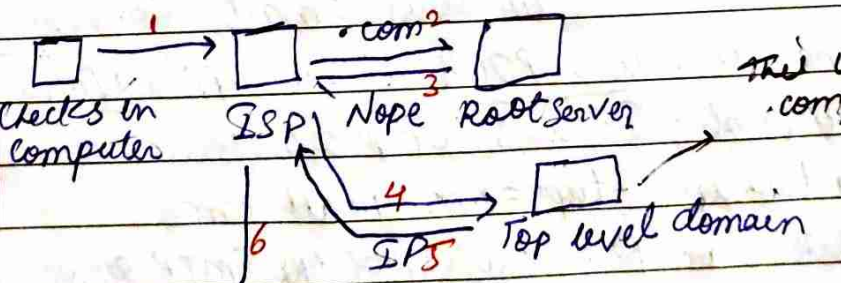
mail · google · com ⇝ Top lvl domain (Root DNS Server)

sub-domain second lvl domain

↳ Root DNS servers : .com, .io, .org

The ICANN (Internet corporation for assigned names and numbers) who register the top level domains.

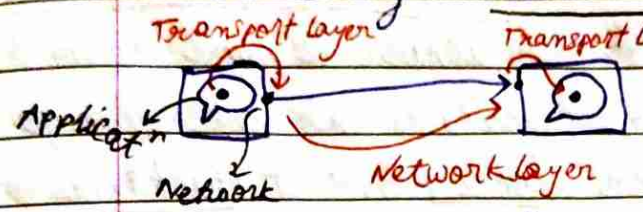
↳ When we visit a site for first time it will store the value of its IP address, that it has found in your local cache. Bcoz we dont want to search it again and again. If not found it fetches the IP in the local DNS server. (your internet service provider) ISP has all the data of websites



We can only rent a Domain name not buy it even from Godaddy it rents from ICANN

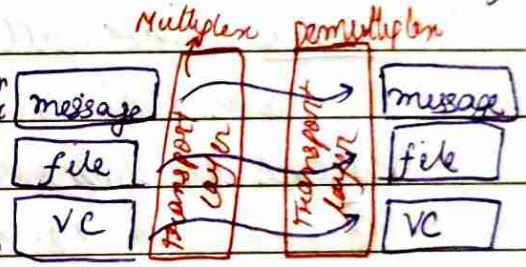
dig google.com ⇝ DNS lookup utility

2. Transport Layer: Is the layer that lies in the devices. Take the information from the network to the application. So when u send message on whatsapp the message from application to the network is done by transport layer. And the data from one host to another is done by Network layer.



So say you are sending 15 files to your friend network layer will send it to your friend but transport layer will decide to which application we need to send these files.

↳ Multiplexing & demultiplexing: we are sending data in a bunch, message, email, video call all together through the network, So how to send these data to the network is called multiplexing and once data is received on friends device we demultiplex the data to send it into various applications.

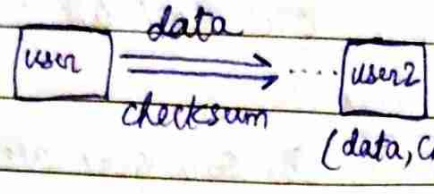


⇒ So whatsapp sends the data to Socket, these Sockets have port number. Data travels in packet, transport layer will attach this Socket port numbers. So it knows from where it has come and where to send.

⇒ Transport layer also takes care of the congestion control, i.e. traffic control. It also has in network layer

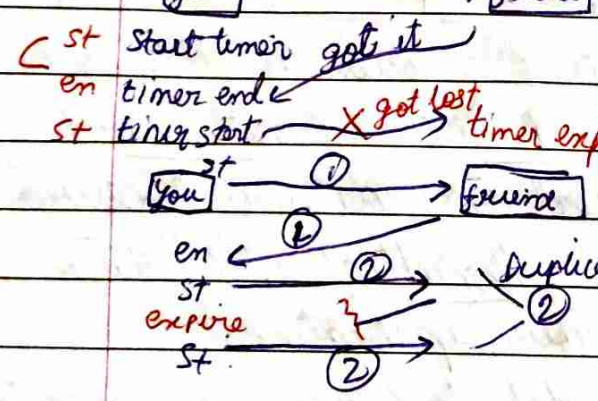
⇒ Congestion control algorithm built in TCP.

↳ **Checksums** - To ensure data sent to your friend reaches completely, there is no loss of data packets and also the data sent is in correct sequence takes care by the transport layer and it uses the concept of checksum



So when we send data a checksum is generated strings which is send with the data. Even the user2 calculates the checksum for the data (when received) with same algorithm. If checksums are same its fine or else something is wrong.

↳ **Timers** : How will I know if the packets that I have sent is recieved by my friend is tracked by timers. So in dig. for second file



the timer expires and we know file never reached called Retransmission timer. So u will send 2 packets duplicate of ② there will be duplicate it is resolved by sequence no

↳ **Transport layer protocols** :

a) **UDP** - User Datagram protocol, here data may or may not be delivered, may change, may not be in order. Connectionless protocol, no connect^n will be established b/w the computers and will still send the data.

⇒ UDP uses checksum to check if it is corrupted but be like it is what it is.

Every data packet will have source & destinat^n port no will be available. Length of the packet/datagram is added, checksum will be added and atv the data

Src port no. 2b	length of datagram 2b	header 8 bytes
Dest. port no. 2b	Checksum 2b	
Data $2^{16}-8 = 65,536$		

Total size $= 2^{16}$ Each packet holds $2^{16}-8$ bytes of data

⇒ Use cases for UDP:- Fast, Videocon conf app, DNS uses UDP coz its fast, gaming.

sudo tcpdump -c 5 fetches 5 data packets and we can view the information of those packets.

b) TCP - Transmission control protocol, it is a transport layer protocol and does the same thing what UDP does but differently.

⇒ Application layer sends a lot of raw data and TCP segments this data i.e divide in chunks, add headers, checksums. It may also collect the data from network layer which might be broken further so it assembles it back and passes to the application layer.

⇒ Congestion control: Handling traffic

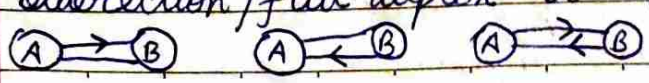
⇒ Takes care of 2 things when data does not arrive and maintains the order of data using the sequence numbers.

↳ Features of TCP:

⇒ Connection oriented and connect^n is established

⇒ Error control, congestion control.

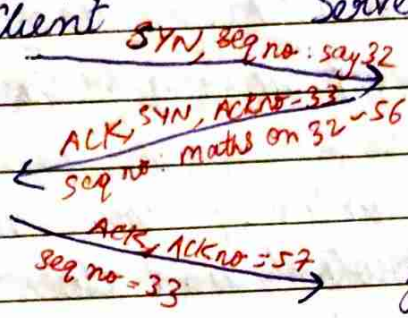
⇒ Bidirection/full duplex: 2 way data can be passed



⇒ One TCP connection can only send to 2 users it cannot send to 10 computers as bidirectional connection has to setup.

⇒ The data packet has the similar header as UDP including all we also add sequence no. and acknowledgement no. as well.

↳ 3 way handshake :



- SYN Synchronist^n flag is sent to establish a connection.

For security seq no. are random. ACK is acknowledgement and gives a math algo on 32 for security.

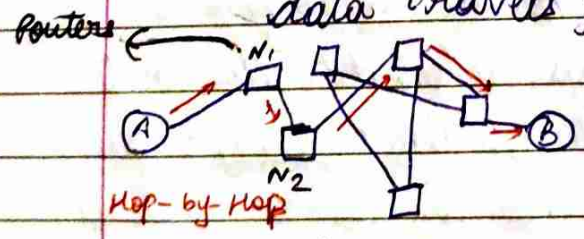
Acknowledgement no. is +1 from previous seq no.

Seq = 32 | ACK = 33 & Seq = math(32) = 56 | ACK = 57, Seq = 33

+1 +1

Now connection is established using 3 way handshake

3- Network layer : Here we work with routers, data travels from one place to another.



Transport : Segments
Network : Data packets
Data link : Frames

So data is sent from Ⓐ to N_1 now N_2 which check it in its routing table and it forward it to next route as it is not for N_1. Forwarding & routing tables live inside the router.

192 · 168 · 2 · 30 → Host ID

network address (Subnet ID) | device address

↳ Control plane: Creates the routing tables, every router is a node and link b/w them is an edge. There are 2 types of routing

⇒ Static Routing which is manually add routes in the routing table.

⇒ Dynamic Routing it evolves when there is a change in network. It uses algo's like Djisktra, Bellman Floyd, Path finding algo.

↳ IP (Internet protocol) it is a network layer protocol. There are version of IPv4, IPv6.
IPv4 → 32 bits number, 4 words it can define a server, client, node or router uniquely defines
IPv6 → 128 bits

8bit [5 . 6 . 9 . 14 Routers holds the info of
[00000101] 8×4=32 bits block of IP address.
So each router which perform hop-by-hop holds the info of block of IP, it is generally the ISP not our routers take part in this.
All the device say in airtel will have same network address. These are known as Subnet ID.
Whenever router will forward a packets it should know the subnet of the destination.
Internet society etc created classes of it
Class of IP address

A	0.0.0.0	–	127.255.255.255
B	128.0.0.0	–	191.255.255.255
C	192.0.0.0	–	223.255.255.255
D	224.0.0.0	–	239.255.255.255
E	240.0.0.0	–	255.255.255.255

↳ Subnet masking it will mask the network part of the IP address and leave for us to use the host part. So in subnet of class E can be 255.255.0.0, I can add any numbers here, → it belongs to class E

↳ Variable length subnet: You can set your own length of the ~~no~~ subnet network

Ex 12.0.0.0/31 → First 31 bits are basically my subnet part.

192.0.1.0/24 → start 192.0.1.0 end can be 192.0.1.1, 192.0.1.2, ⋮, 192.0.1.255

i.e $2^{(32-24)} = 2^8$ bits are remaining to form IP.

↳ IETF assigns the IP based on the regions to minimise the hops. There are some reserved addresses

Ex 127.0.0.0/8 → reserves the first 8 bits prefix

localhost: 127.0.0.1 also known as loopback addresses, allows u to contact the same processes for local environment.

↳ Packets: Here the header is of 20 bytes

IP v, length, identificat^n no, flags, protocols, checksum, addresses, TTL, etc. → time to live

TTL is basically say ttl=60 means after 60 hops if the packet doesn't reach the destination the packet will be dropped.

↳ IPv4 v/s IPv6

IPv4: $2^{32} \approx 4.3$ billion unique IP address

IPv6: $2^{32\times4} = 2^{128} \approx 3.4\times10^{38}$ IP addresses.

⇒ Cons : Not backward compatible, as devices on IPv4 cannot access websites or servers with IPv6. It requires effort to shift ISP, hardware

⇒ How is IPv6 represented it has 8 numbers and every number is a hexadecimal digit represents 16 bit part of the address.

a : a : a : a : a : a : a : a

Hexadecimal (16 bit string)

Ex ABFE : F001 : 3210 : 9182 : 0 : 0 : 1 : 3

Can do the same representat^n of subnet here as well

ABFE : F001 : 0 : 9182 :: /60 → prefix subnet

set of zeroes

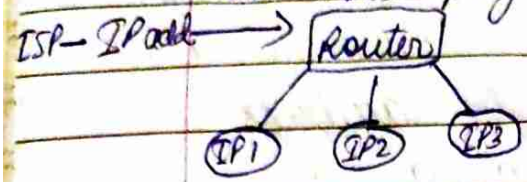
↳ Middle Boxes: We have seen computer and routers that interact with each other, there is this middle boxes as well which interacts with the network packets. They can be in the network layer or transport layer. for ex

⇒ Firewall: Fitters out IP Packet based on various rules, for trusted networks. Fitlers out based on the address, packets, port numbers, flags, protocols.

⇒ It is in 2 types stateless and stateful firewalls Stateful is more efficient as it store cached packets and allows similar packets to pass.

↳ NAT (Network address translation) is another type of middlebox method which maps an IP add into another by modifying network address information in the IP header of packets.

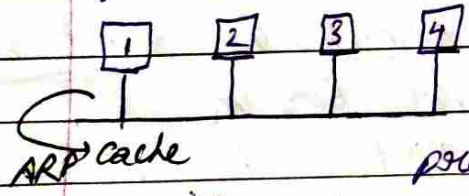
4. Data link Layer : Data packets that we receive from the network layer, data link layer is responsible to send these packets to the physical layer.



Router assigns IP address to the devices connected to this router using DHCP. It ~~adds~~ has a subnet which allocates new IP address

Every DHCP server will have some pool of IP addresses and it assigns it from that.

↳ Many devices are connected in LAN. But at the data link layer they communicate with each other using the data link layer address. You can manually assign it or it can also happen automatically.



Say 1 wants to connect to 4. It will check in cache if not present, it will check in ARP cache as it sends frame to all the devices connected to this LAN. Frame contains data link layer address of sender and IP address of destination i.e 4, with this it updates its cache known as ARP (Address resolution protocol). This data link layer address is known as the mac addresses of your device when we talk about ethernets. So mac address is not of device but of component like wifi, bluetooth. Router attaches the correct IP address to the data packets ensuring they go to the correct destination.